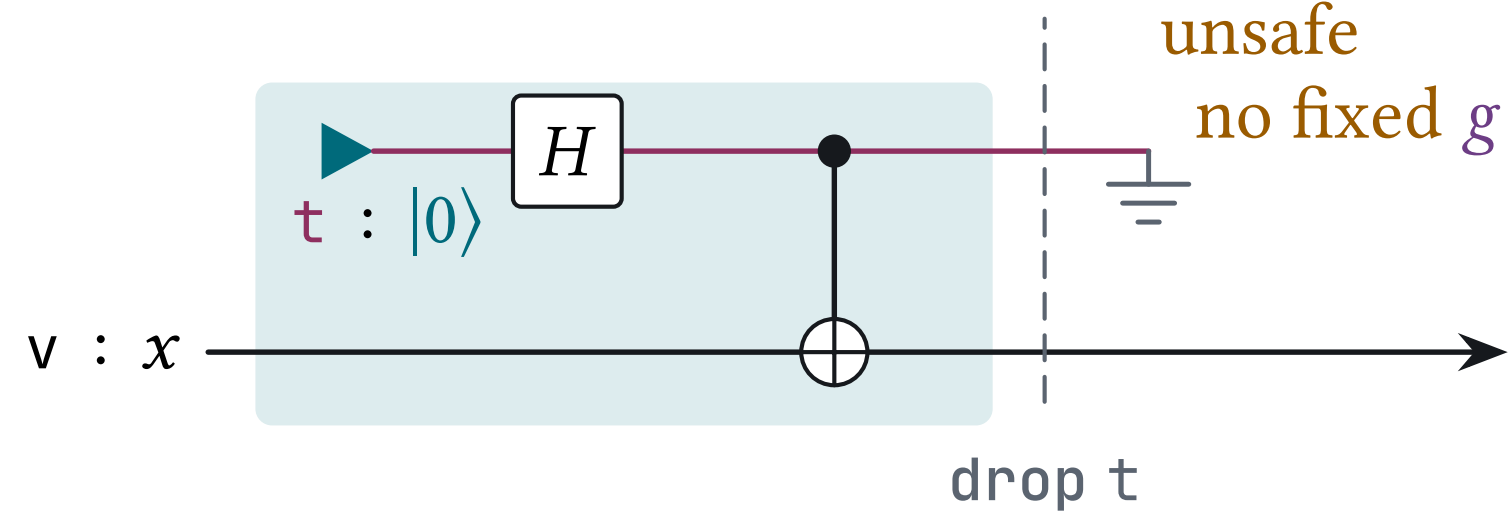$$K^{(b)}x = (|0\rangle \otimes x + |1\rangle \otimes Xx)/\sqrt{2}$$